

MVA Research Internship
Report

Supervised Classification for Diffusion Processes

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Chapter 1

Proofs

Construction of the Bayes classifier

Proof. Let's prove $g^* \in \operatorname{argmin}_{g \in \mathcal{G}} \mathcal{R}(g) \iff g^*(X) \in \operatorname{argmax}_{i \in \mathcal{Y}} \pi_i^*(X)$, where $\pi_i^*(X) := \mathbb{P}(Y = i|X)$.

We have the following

$$\begin{aligned}
 g^* \in \operatorname{argmin}_{g \in \mathcal{G}} \mathcal{R}(g) &\iff g^* \in \operatorname{argmin}_{g \in \mathcal{G}} \mathbb{E} [\mathbb{1}_{\{g(X) \neq Y\}}] \\
 &\iff g^* \in \operatorname{argmin}_{g \in \mathcal{G}} \mathbb{E} \left[\sum_{i=1}^K \mathbb{1}_{\{g(X) \neq i\}} \mathbb{1}_{\{Y=i\}} \right] \\
 &\iff g^* \in \operatorname{argmin}_{g \in \mathcal{G}} \mathbb{E} \left[\mathbb{E} \left[\sum_{i=1}^K \mathbb{1}_{\{g(X) \neq i\}} \mathbb{1}_{\{Y=i\}} | X \right] \right] \\
 &\iff g^* \in \operatorname{argmin}_{g \in \mathcal{G}} \mathbb{E} \left[\sum_{i=1}^K \mathbb{1}_{\{g(X) \neq i\}} \pi_i^*(X) \right] \\
 &\iff g^* \in \operatorname{argmin}_{g \in \mathcal{G}} \mathbb{E} \left[\sum_{i=1}^K \sum_{k=1, k \neq i}^K \mathbb{1}_{\{g(X)=k\}} \pi_i^*(X) \right] \\
 &\iff g^* \in \operatorname{argmin}_{g \in \mathcal{G}} \mathbb{E} \left[\sum_{k=1}^K \mathbb{1}_{\{g(X)=k\}} \sum_{i=1, i \neq k}^K \pi_i^*(X) \right]
 \end{aligned}$$

Hence, finding the $\operatorname{argmin} g \in \mathcal{G}$ is equivalent to minimize the function inside the expectation. We then have

$$\begin{aligned}
 g^*(X) = k &\iff \forall j \in \mathcal{Y}, j \neq k, \sum_{i=1, i \neq k} \pi_i^*(X) \leq \sum_{i=1, i \neq j} \pi_i^*(X) \\
 &\iff \forall j \in \mathcal{Y}, j \neq k, \pi_j^*(X) + \sum_{i=1, i \neq k, j} \pi_i^*(X) \leq \pi_k^*(X) + \sum_{i=1, i \neq k, j} \pi_i^*(X) \\
 &\iff \forall j \in \mathcal{Y}, j \neq k, \pi_j^*(X) \leq \pi_k^*(X) \\
 &\iff k \in \operatorname{argmax}_{i \in \mathcal{Y}} \pi_i^*(X)
 \end{aligned}$$

□

Proof of proposition 2.3

Proof. The probability measure $\mathbb{P}$ can be decomposed in $\mathbb{P} = \sum_{i=1}^K p_i \mathbb{P}_i$, with $\mathbb{P}_i = \mathbb{P}(\cdot | Y = i)$. Let's construct a brownian motion $\widetilde{W}$, under a probability $\mathbb{P}_0$, such that $(X_t)_{t \geq 0}$ is solution of $dX_t = \sigma(X_t) d\widetilde{W}_t$, in order to represent the process X "independently" of the labels.

Therefore, we have :

$$\begin{aligned} dX_t &= \sigma(X_t) d\widetilde{W}_t = \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} b_i(X_t) dt + \sigma(X_t) dW_t \\ \implies d\widetilde{W}_t &= dW_t + \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \frac{b_i}{\sigma}(X_t) dt \\ \implies \widetilde{W}_t &= W_t + \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i}{\sigma}(X_t) dt \\ \implies \widetilde{W}_t &= W_t + \langle W, \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i}{\sigma}(X_s) dW_s \rangle_t . \end{aligned}$$

As

$$dX_t = \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} b_i(X_t) ds + \sigma(X_t) dW_t \implies dW_t = \frac{dX_t}{\sigma(X_t)} - \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \frac{b_i}{\sigma}(X_t) ds$$

We have

$$\begin{aligned} \langle W, \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i}{\sigma}(X_s) dW_s \rangle_t &= \langle W, \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i}{\sigma^2}(X_s) dX_s - \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i^2}{\sigma^2}(X_s) ds \rangle_t \\ &= \langle W, \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i}{\sigma^2}(X_s) dX_s \rangle_t - \langle W, \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i^2}{\sigma^2}(X_s) ds \rangle_t \\ &= \langle W, \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i}{\sigma^2}(X_s) dX_s \rangle_t . \end{aligned}$$

Finally, we have

$$\widetilde{W}_t = W_t - \langle W, \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int -\frac{b_i}{\sigma^2}(X_s) dX_s \rangle_t .$$

Let's define the process $L = (\sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int_0^t -\frac{b_i}{\sigma^2}(X_s) dX_s)_{t \in [0,1]}$. We have

$$\begin{aligned} \langle L, L \rangle_t &= \langle \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i}{\sigma}(X_s) dW_s, \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i}{\sigma}(X_s) dW_s \rangle_t \\ &= \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i^2}{\sigma^2}(X_s) ds . \end{aligned}$$

Then, let's define the process

$$D = (\exp(L_t - \frac{1}{2} \langle L, L \rangle_t))_{t \in [0,1]} = (\exp \left(\sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \left(\int_0^t \frac{-b_i}{\sigma^2}(X_s) dX_s - \int_0^t \frac{b_i^2}{2\sigma^2}(X_s) ds \right) \right))_{t \in [0,1]}$$

And the probability $\mathbb{P}_0$ such that :

$$\forall A \in F_X^1, \mathbb{P}_0(A) = \mathbb{E}_{\mathbb{P}} [\mathbb{1}_{\{A\}} D_1]$$

Hence, by assumption 2.2 (Novikov's condition), D is a martingale under $\mathbb{P}$ and $\mathbb{P}_0 \ll \mathbb{P}$, with : $\forall t \in [0, 1], D_t = \frac{d\mathbb{P}_0}{d\mathbb{P}}|_{F_X^t}$.

Applying Girsanov's Theorem, we have that $\widetilde{W}$ is a brownian motion under $\mathbb{P}_0$.

On the other hand, we have :

$$\begin{aligned} \forall t \in [0, 1], D_t &= \exp \left(\sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \left(\int_0^t \frac{-b_i}{\sigma^2} (X_s) dX_s - \int_0^t \frac{b_i^2}{2\sigma^2} (X_s) ds \right) \right) \\ \iff \forall t \in [0, 1], D_t &= \prod_{i=1}^K \mathbb{1}_{\{Y=i\}} \exp \left(\int_0^t \frac{-b_i}{\sigma^2} (X_s) dX_s - \int_0^t \frac{b_i^2}{2\sigma^2} (X_s) ds \right) \\ \iff \forall t \in [0, 1], D_t &= \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \exp \left(\int_0^t \frac{-b_i}{\sigma^2} (X_s) dX_s - \int_0^t \frac{b_i^2}{2\sigma^2} (X_s) ds \right) \text{ (disjoint sets)} \end{aligned}$$

Hence,

$$\begin{aligned} \forall A \in F_X^1, \mathbb{P}_0(A) &= \mathbb{E}_{\mathbb{P}} [\mathbb{1}_{\{A\}} D_1] \\ &= \sum_{i=1}^K p_i \mathbb{E}_{\mathbb{P}_i} [\mathbb{1}_{\{A\}} D_1] \\ &= \sum_{i=1}^K p_i \mathbb{E}_{\mathbb{P}_i} \left[\mathbb{1}_{\{A\}} \exp \left(\int_0^1 \frac{-b_i}{\sigma^2} (X_s) dX_s - \int_0^1 \frac{b_i^2}{2\sigma^2} (X_s) ds \right) \right]. \end{aligned}$$

We can finally deduce

$$\forall i \in \{1, \dots, K\}, \frac{d\mathbb{P}_0|_{\mathcal{F}_t^X}}{d\mathbb{P}_i|_{\mathcal{F}_t^X}} = \exp \left(\int_0^1 \frac{-b_i}{\sigma^2} (X_s) dX_s - \int_0^1 \frac{b_i^2}{2\sigma^2} (X_s) ds \right)$$

Notice that we constructed a new brownian motion $\widetilde{W}$ from W , under a new mesure $\mathbb{P}_0$ from $\mathbb{P}$. This led us to compute the quantity $\frac{d\mathbb{P}_0|_{\mathcal{F}_t^X}}{d\mathbb{P}_i|_{\mathcal{F}_t^X}}$. If we want to compute the reversed quantity, i.e $\frac{d\mathbb{P}_i|_{\mathcal{F}_t^X}}{d\mathbb{P}_0|_{\mathcal{F}_t^X}}$, we

have to follow the same reasoning, but now constructing W from $\widetilde{W}$, and $\mathbb{P}$ from $\mathbb{P}_0$.

Remark that, using all notations defined previously, we already have

$$\begin{aligned} d\widetilde{W}_t &= dW_t + \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \frac{b_i}{\sigma} (X_t) dt \\ \implies dW_t &= d\widetilde{W}_t - \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \frac{b_i}{\sigma} (X_t) dt \\ \implies W_t &= \widetilde{W}_t - \sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \int \frac{b_i}{\sigma} (X_t) dt \\ \implies W_t &= \widetilde{W}_t - \langle \widetilde{W}, -L \rangle_t \end{aligned}$$

So, we can apply the exact same reasoning using $-L$ instead of L , and adapting

$$D = (\exp(-L_t - \frac{1}{2} \langle L, L \rangle_t))_{t \in [0,1]} = (\exp \left(\sum_{i=1}^K \mathbb{1}_{\{Y=i\}} \left(\int_0^t \frac{b_i}{\sigma^2} (X_s) dX_s - \int_0^t \frac{b_i^2}{2\sigma^2} (X_s) ds \right) \right))_{t \in [0,1]}$$

And then

$$\forall i \in \{1, \dots, K\}, \Phi_t^i := \frac{d\mathbb{P}_i|_{\mathcal{F}_t^X}}{d\mathbb{P}_0|_{\mathcal{F}_t^X}} = \exp \left(\int_0^t \frac{b_i}{\sigma^2} (X_s) dX_s - \int_0^t \frac{b_i^2}{2\sigma^2} (X_s) ds \right)$$

Thus, for $t \in [0, 1]$, we have that

$$\mathbb{P}_{\mathcal{F}_t^X} = \sum_{i=1}^K p_i \mathbb{P}_i|_{\mathcal{F}_t^X} = \sum_{i=1}^K p_i \Phi_t^i \mathbb{P}_0|_{\mathcal{F}_t^X}$$

Denote

$$\forall i \in \{1, \dots, K\}, \Psi_t^i := \frac{d\mathbb{P}_i|_{\mathcal{F}_t^X}}{d\mathbb{P}|_{\mathcal{F}_t^X}} = \frac{\Phi_t^i d\mathbb{P}_0|_{\mathcal{F}_t^X}}{\sum_{j=1}^K p_j \Phi_t^j d\mathbb{P}_0|_{\mathcal{F}_t^X}} = \frac{\Phi_t^i}{\sum_{j=1}^K p_j \Phi_t^j}$$

Moreover, let $h : \{1, \dots, K\} \rightarrow \mathbb{R}$ a bounded function and Z an $\mathcal{F}_t^X$ -measurable bounded random variable. We have that

$$\begin{aligned} \mathbb{E}[h(Y)Z] &= \mathbb{E}[h(Y)\mathbb{E}[Z | \sigma(Y)]] = \mathbb{E}\left[\sum_{i=1}^K h(i) \mathbb{1}_{\{Y=i\}} \mathbb{E}_i[Z]\right] \\ &= \mathbb{E}\left[\sum_{i=1}^K h(i) \mathbb{1}_{\{Y=i\}} \mathbb{E}[\Psi_t^i Z]\right] = \sum_{i=1}^K p_i h(i) \mathbb{E}[\Psi_t^i Z] \\ &= \mathbb{E}\left[\left(\sum_{i=1}^K h(i) p_i \Psi_t^i\right) Z\right] \end{aligned}$$

giving

$$\mathbb{E}[h(Y) | \mathcal{F}_t^X] = \sum_{i=1}^K p_i h(i) \Psi_t^i \quad \mathbb{P} - a.s$$

Using definition (7.5) we obtain

$$\mathbb{E}[h(Y) | \mathcal{F}_t^X] = \frac{\sum_{i=1}^K h(i) p_i \Phi_t^i}{\sum_{i=1}^K p_i \Phi_t^i}$$

Thus, for $h(x) = \mathbb{1}_{\{i\}}(x)$ we obtain the expected result. □